14/10/26

Signal Processing

- Recap: LCCDE ✓
- System Analysis: Example: $H(z) = (1+z^{-1})^2 / (1-\frac{1}{2}z^{-1})\cdot(1+\frac{5}{4}z^{-1})$ ✓
  - Stability and Causality; $y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$ ✓
  - Inverse Systems; $H(z) = (1-0.5z^{-1})/(1-0.9z^{-1})$; $H(z) = (z^{-1}-0.5)/(1-0.9z^{-1})$ ✓
  - Stability and causality for forward and inverse systems ✓
  - Minimum-phase systems.
- LCCDE represents linear systems when the initial condition is that of rest.
- LCCDE:

$$\sum_{k=0}^{M} a_k\, y[n-k] = \sum_{k=0}^{N} b_k\, x[n-k] \quad -(1)$$

$$\Rightarrow \sum_{k=0}^{M} a_k\, Y(z)\, z^{-k} = \sum_{k=0}^{N} b_k\, X(z)\, z^{-k} \quad -(2)$$

Assuming initial rest condition,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k \cdot z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} \quad -(3)$$

$$H(z) = \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=0}^{M} (1 - c_k z^{-1})}{\prod_{k=0}^{N} (1 - d_k z^{-1})} \quad -(4)$$

- LCCDE example: $y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$

$$Y(z)\left(1 - \frac{5}{2}z^{-1} + z^{-2}\right) = X(z)$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]\, z^{-n} \qquad Y(z)\left(1-\frac{1}{2}z^{-1}\right)\left(1-2z^{-1}\right) = X(z)$$

$$\Rightarrow H(z) = \frac{1}{\left(1-\frac{1}{2}z^{-1}\right)\left(1-2z^{-1}\right)}$$

Three pole Regions of convergence:

R_1: $|z| < \frac{1}{2}$. Not stable since ROC does not contain $|z|=1$

Not causal since ROC is inside the smallest pole.

Left-sided sequence.

R_2: $\frac{1}{2} < |z| < 2$: stable since $|z|=1$ is in the ROC.

Non-causal since the ROC is a disc.

Two-sided? Yes.

R_3: $|z| > 2$. Not-stable

causal

Right-sided.

- Inverse systems: Given $H(z) = \left(\frac{b_0}{a}\right) \cdot \frac{\prod_{k=0}^{N} (1 - c_k z^{-1})}{\prod_{k=0}^{M} (1 - d_k z^{-1})}$

  - $(1 - c_k z^{-1})$ Contributes a zero at c_k at a pole at 0.
  - $(1 - d_k z^{-1})$ Contributes a pole at d_k and a zero at 0.

- We want to find $H_{inv}(z)$ such that

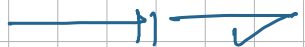
$$H_{inv}(z)\, H(z) = 1$$

$$\text{i.e., } H_{inv}(z) = \frac{1}{H(z)}$$

- For the combined system to be valid, the ROC of $H(z)$ and $H_{inv}(z)$ must have a non-zero overlap.

- The poles of $H(z)$ become the zeros of $H_{inv}(z)$
- The zeros of $H(z)$ become the poles of $H_{inv}(z)$.

Example 1: $H(z) = \frac{1-0.5z^{-1}}{1-0.9z^{-1}}$; $|z| > 0.9$ ✓ Stable + Causal

- $H_{inv}(z) = \frac{1-0.9z^{-1}}{1-0.5z^{-1}}$; $|z| > 0.5$ ✓ 
- $H'_{inv}(z) = \frac{1-0.9z^{-1}}{1-0.5z^{-1}}$; $|z| < 0.5$; non causal, unstable ✓ Not a valid inverse.
- $h_{inv}[n] = (0.5)^n u[n] - 0.9(0.5)^{n-1} u[n-1]$
- $h'_{inv}[n] = -(0.5)^n u[-n-1] + (0.9)(0.5)^{n-1} u[-n-2]$ ⇐ ?
- $h[n] = (0.9)^n u[n] - 0.5 \cdot (0.9)^{n-1} u[n-1]$

- Minimum phase system: An LTI system with poles and zeros inside the unit circle.